



CODE_TYMPAN® SOFTWARE FOR CALCULATING INDUSTRIAL NOISE IN THE ENVIRONMENT

Thibaud Thénint, EDF DTG

Noise Modelling Days 2025

DTG



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- ☐ ▪ Accessibilité très limitée (confidentiel)

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1. INDUSTRIAL CONTEXT: ACOUSTICS IN EDF

■ Multiple stakes

□ Health and well-being

- Professional diseases, deafness, impacts on health and sleep of local residents

□ Technics

- Equipment noise: change = problem? => mechanisms of locks on Rhin river, watertightness of spherical valves
- Safety => ice detection in flow discharge pipes
- Performance/maintenance => acoustics enclosure for engines/turbines, ...

□ Regulation

- “Code du travail” for EDF employees and subcontractors
- Different norms/laws for environmental impact (nuclear, thermal, hydro, wind turbine, ...)

□ Image

- Acceptability of our installations

□ Finance

- Fines if non-respect of the regulatory norm and complaints
- High sound-proofing costs (~1 M€)

1. INDUSTRIAL CONTEXT: ACOUSTICS IN EDF

- Very different installations



- And more: mechanical workshops, electrical transformers, ...
- And also: construction and deconstruction sites

1. INDUSTRIAL CONTEXT: CALCULATION SOFTWARE POLICY

■ Code policy

- EDF develops its own simulation tools for its specific needs
- Regulatory authorities want proofs of mastering of the simulation tools

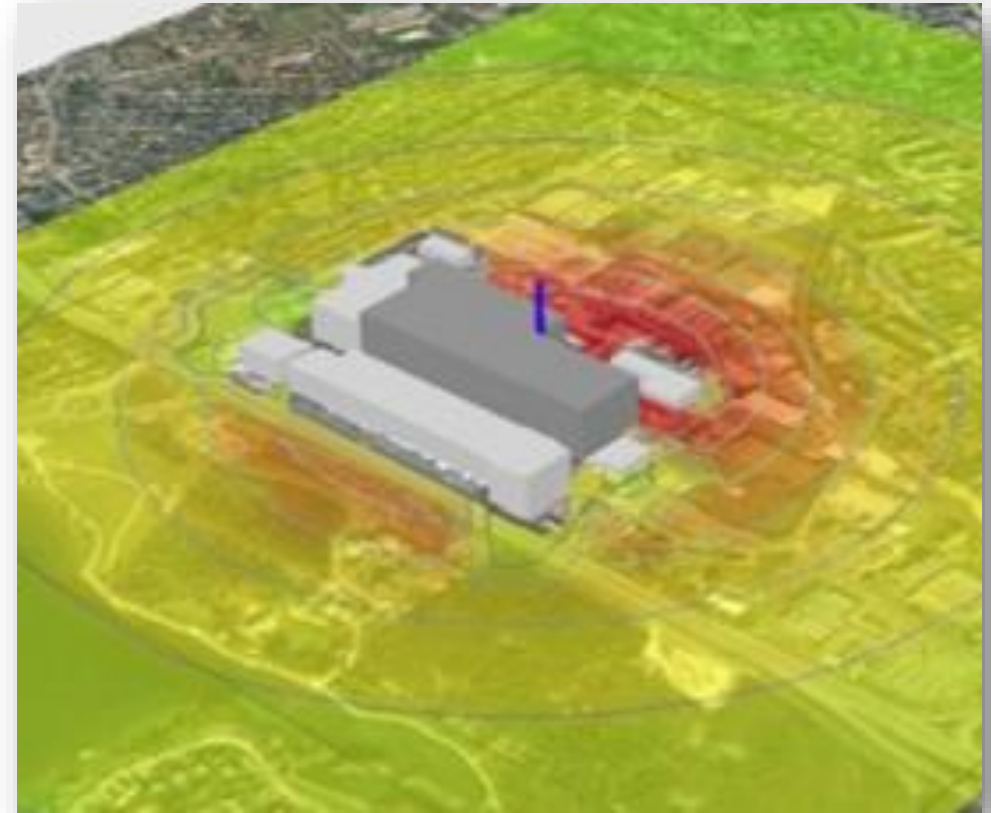
■ Open source

- Increasing the number of external users, and thus the “commercial visibility”
- A “third party” validation and acknowledgement, enhancing credibility of our simulations
- Simplified procedures with external developers

- We sell nothing: we’re not another competitor in the software market
- Counterpart: we can’t guarantee help or support

2. **CODE TYMPAN** DESCRIPTION

- **Open-source software for acoustic propagation in complex 3D scenes to assess and predict the noise impact of industrial sources on the environment**
- **Engineering solver**
 - Propagation based on ISO 9613-2 standard
 - Atmospheric attenuation in line with ISO 9613-1 standard
 - Ground effect attenuation based on ground impedance
- **Main uses:**
 - assess the regulatory compliance of industrial facilities
 - technical arguments and decision support
 - reduce soundproofing costs



2. CODE TYMPAN DESCRIPTION

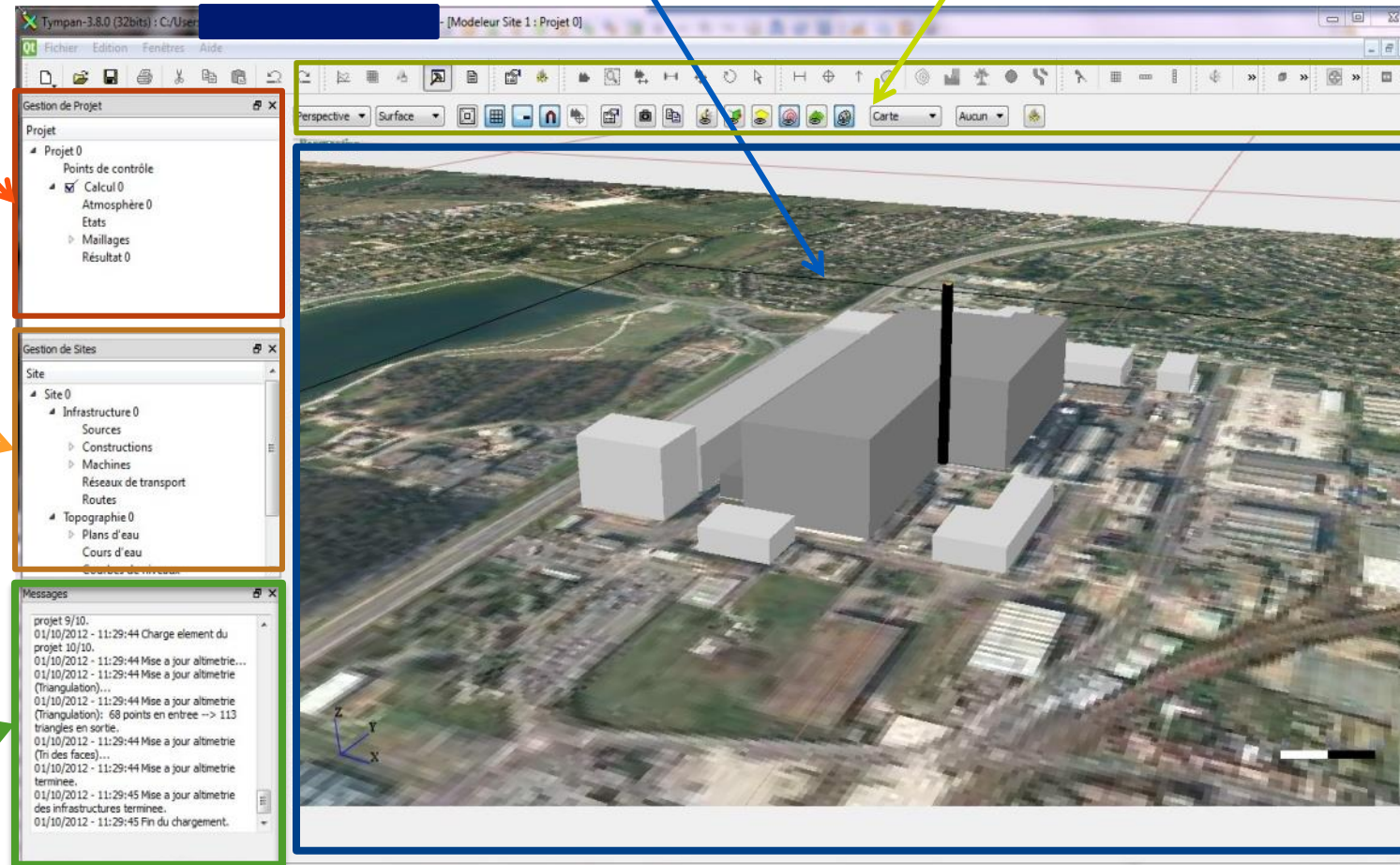
Project
(receivers and
options)

Sites
(machines,
buildings, level
curves, ...)

Messages

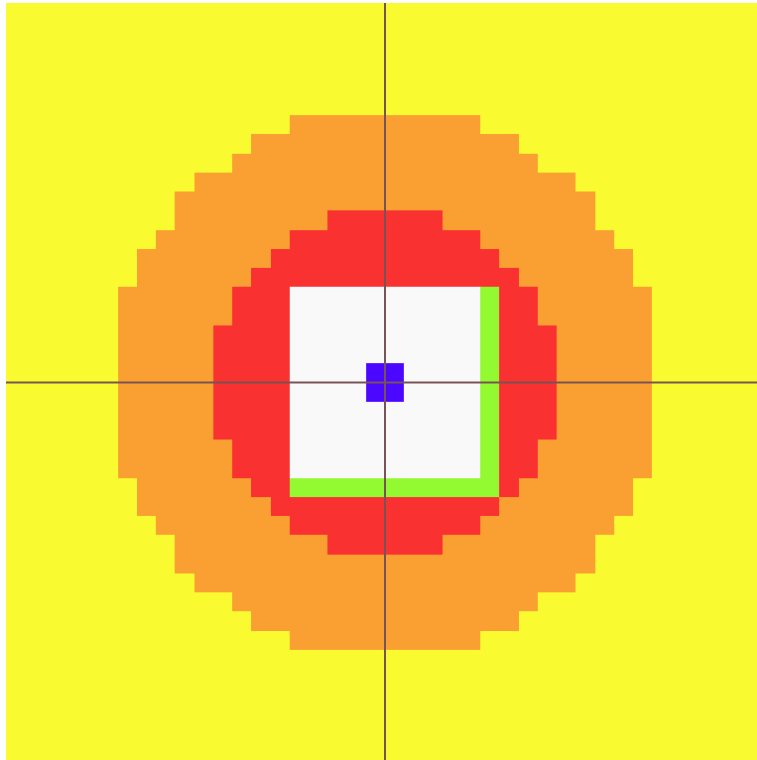
3D modeller

Modeller toolbox

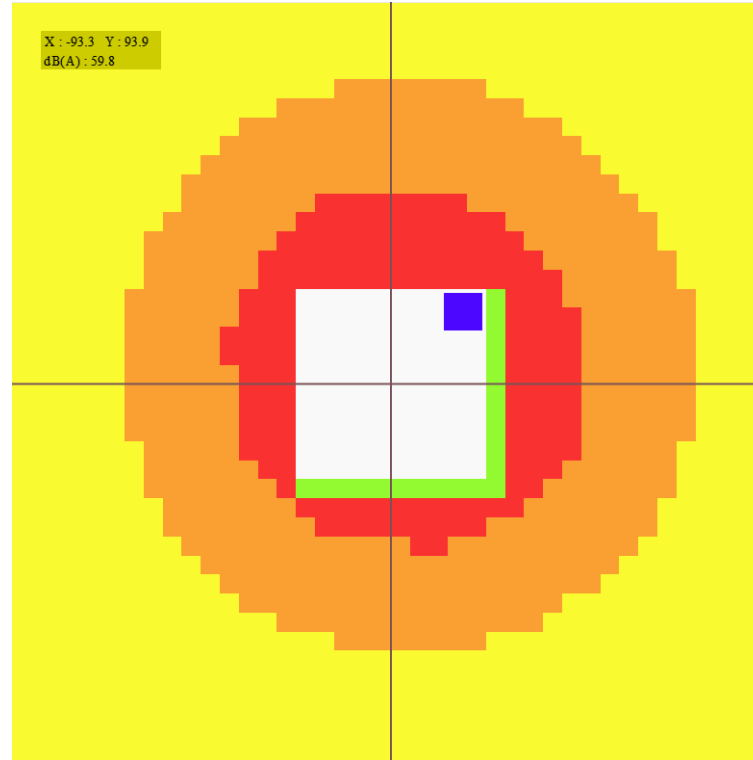


2. CODE TYMPAN DESCRIPTION

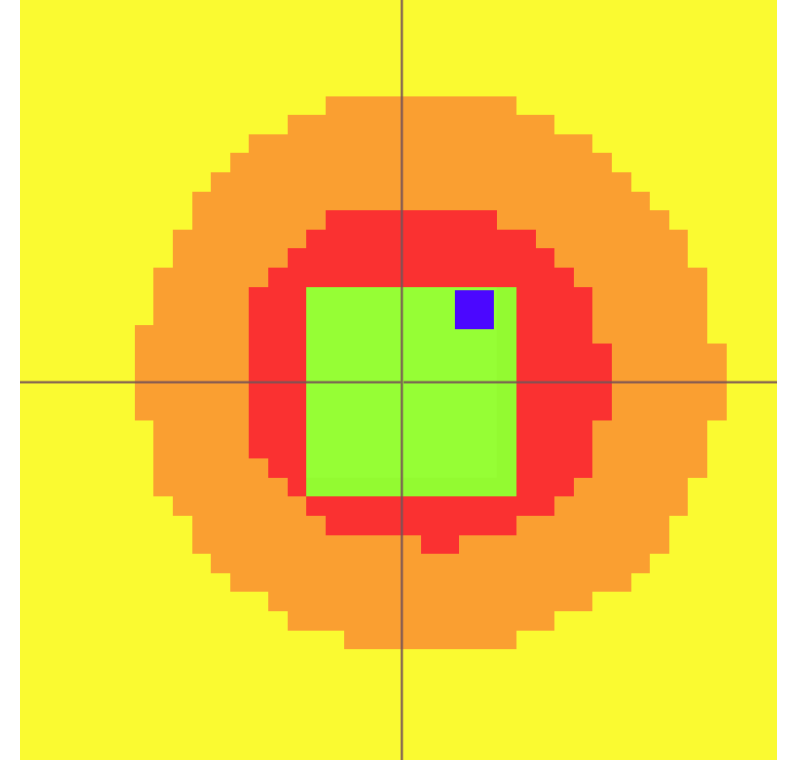
- Industrial sources => machines can be put in buildings + point sources are created



center



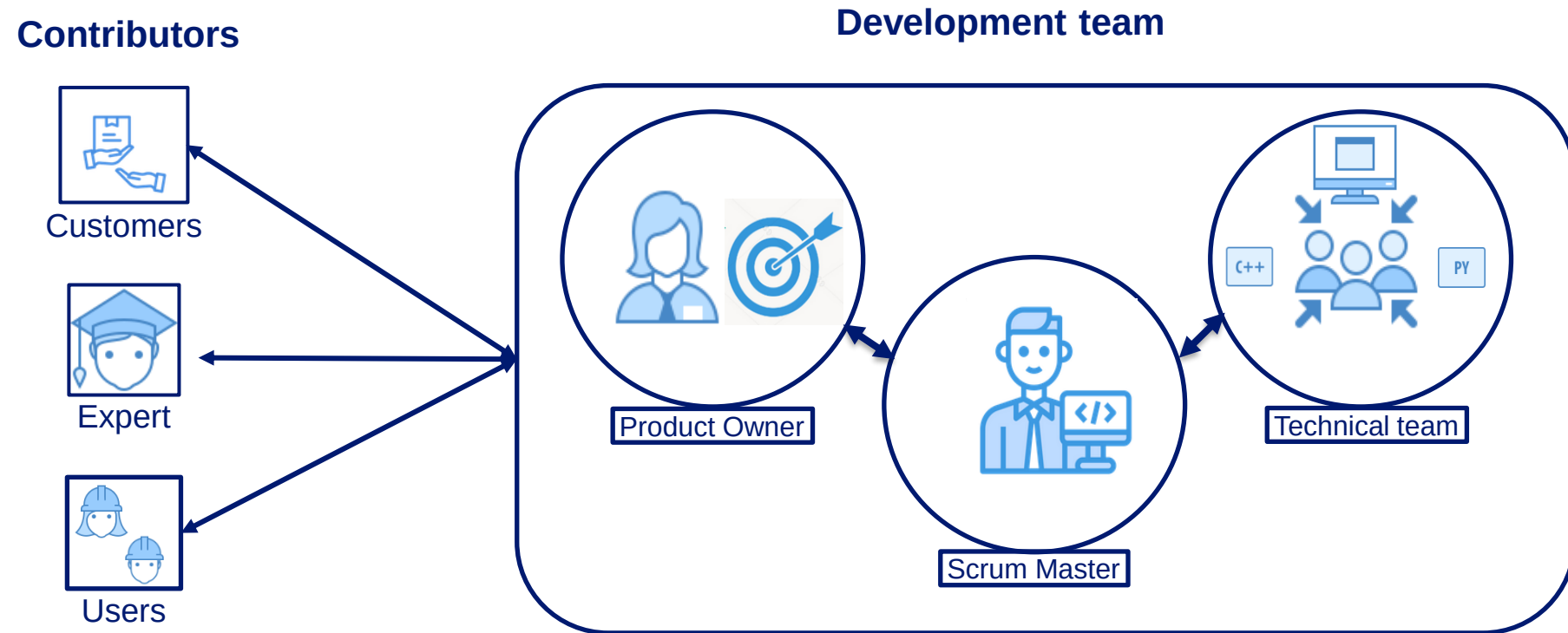
corner



corner
1 radiating face

2. CODE TYMPAN DESCRIPTION

- Agile organisation of development team

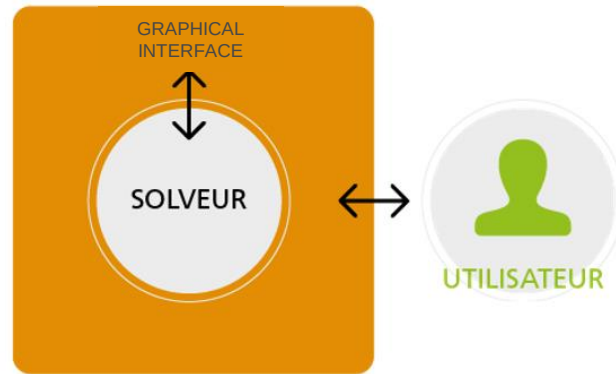


2. NEW FUNCTIONALITIES

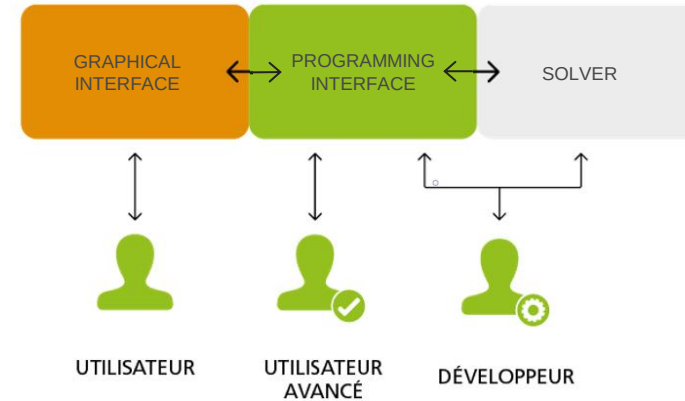
- Bugfixes and ergonomics improvements
- New solver: strict implementation of the ISO 9613
- Review of the user manual + partial translation in English (to be published)
- Automatization of physical tests
- Library obsolescence: Qt5 to Qt6 analysis, VS 2019 to 2022 done

- Open code with business modules

Version 3.x



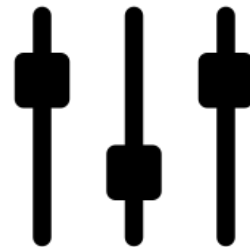
Version 4.x



New tools



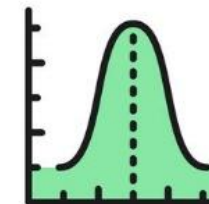
Toolbox



TYKAL



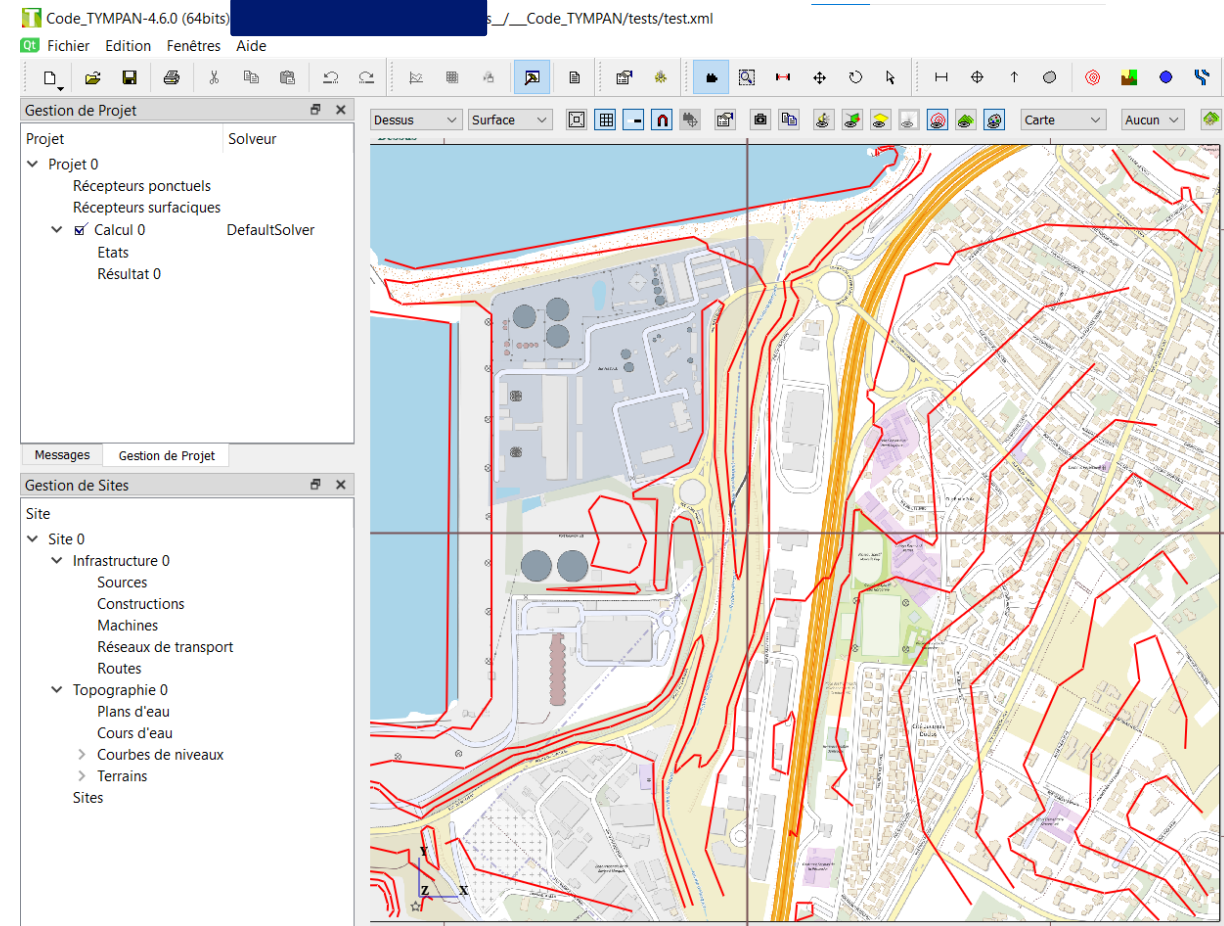
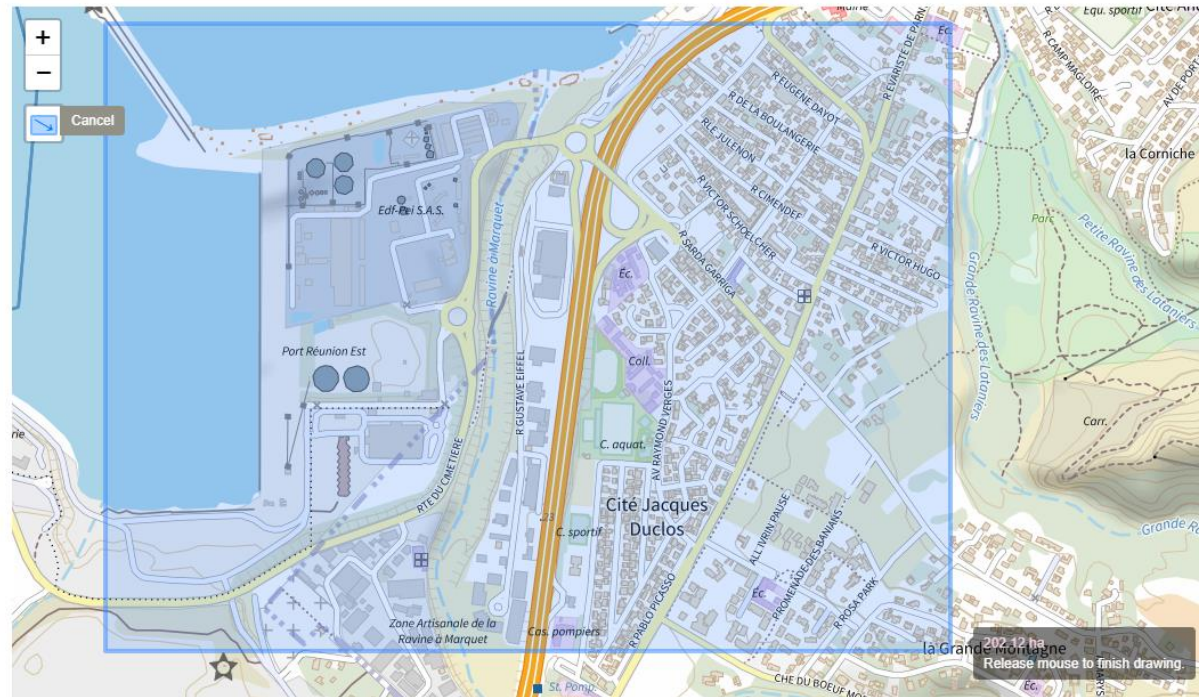
OCAPY



OCP PIPE

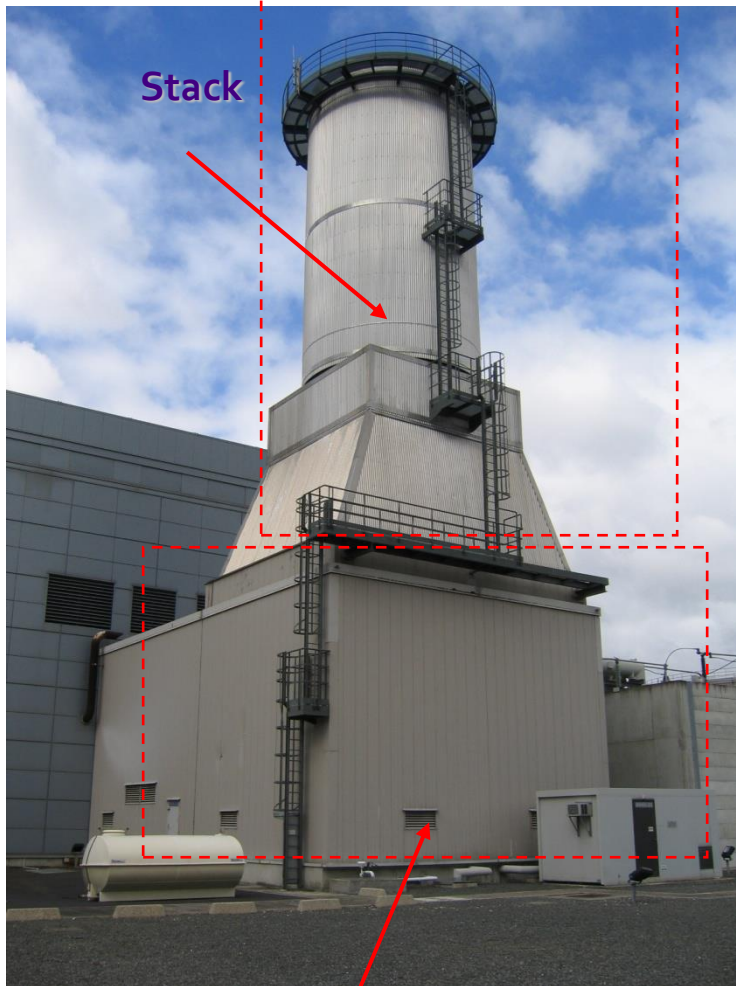
■ Geo-referenced projects (France only)

Sélection de la zone

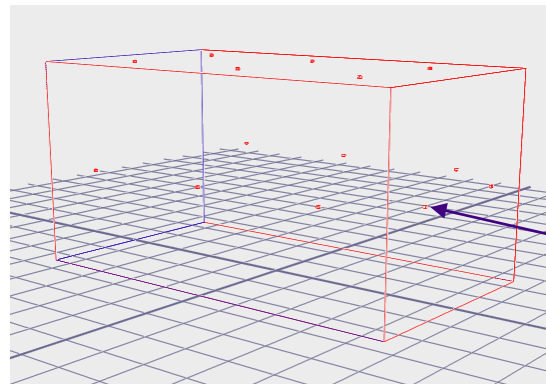


3. REAL USE CASES

- Preliminary designs to assess specifications



Duct



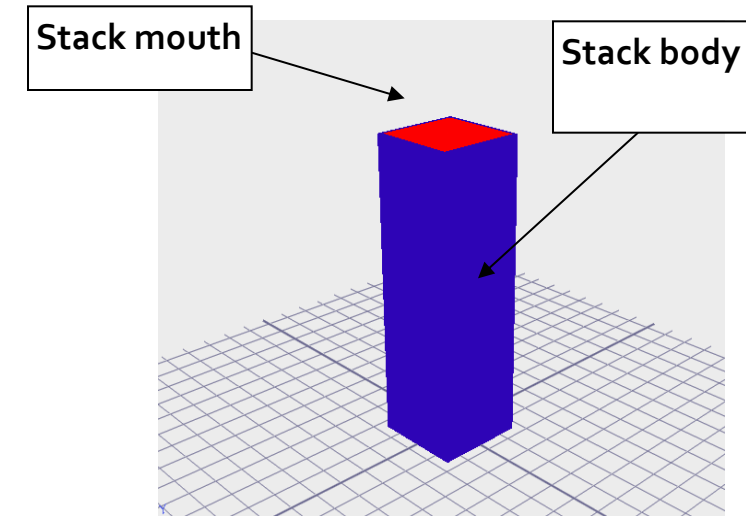
Duct

Acoustic power:

$L_w(\text{stack mouth}) = 97.7 \text{ dB(A)}$

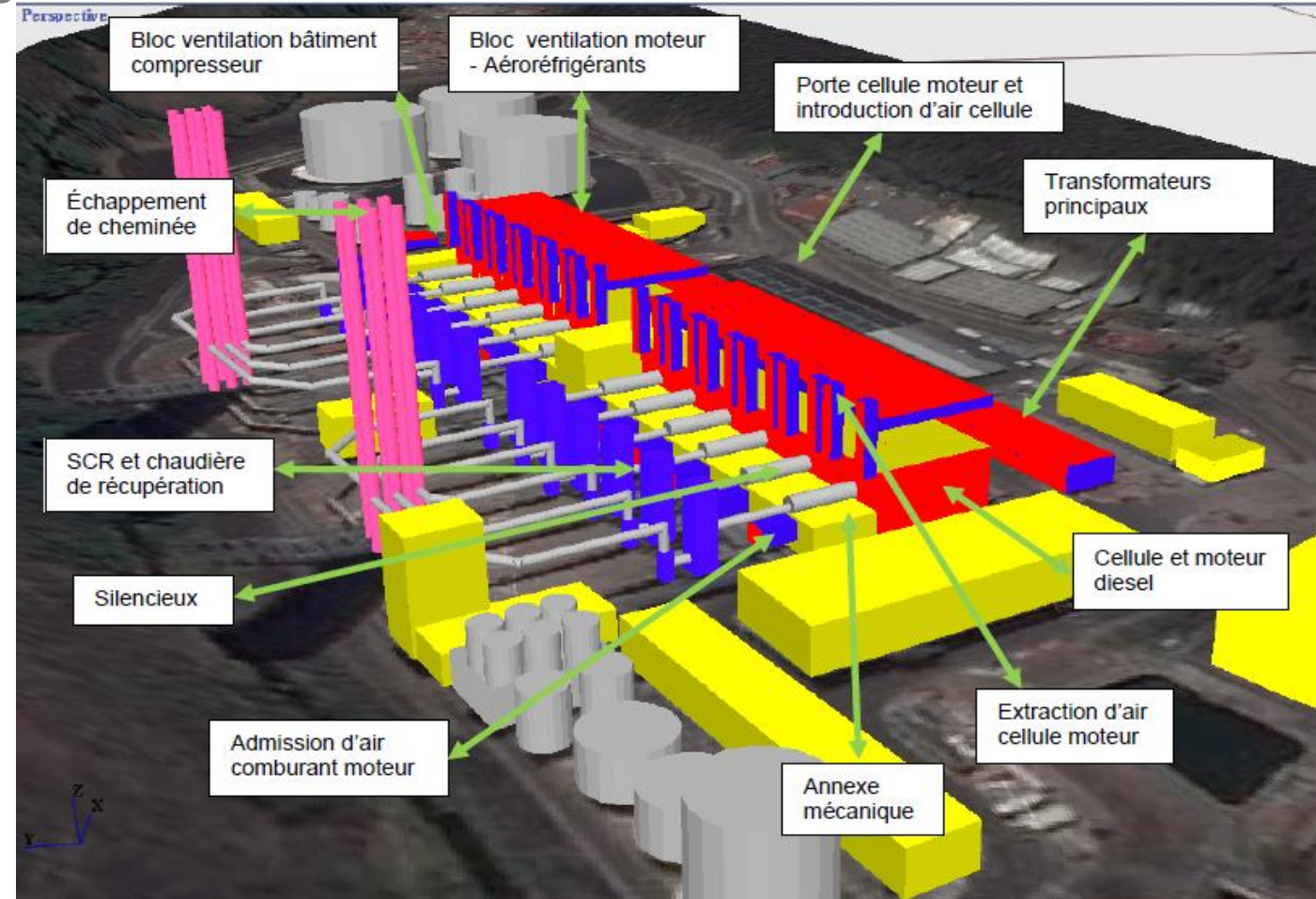
$L_w(\text{stack body}) = 89 \text{ dB(A)}$

$L_w(\text{duct}) = 96.5 \text{ ou } 109 \text{ dB(A)}$



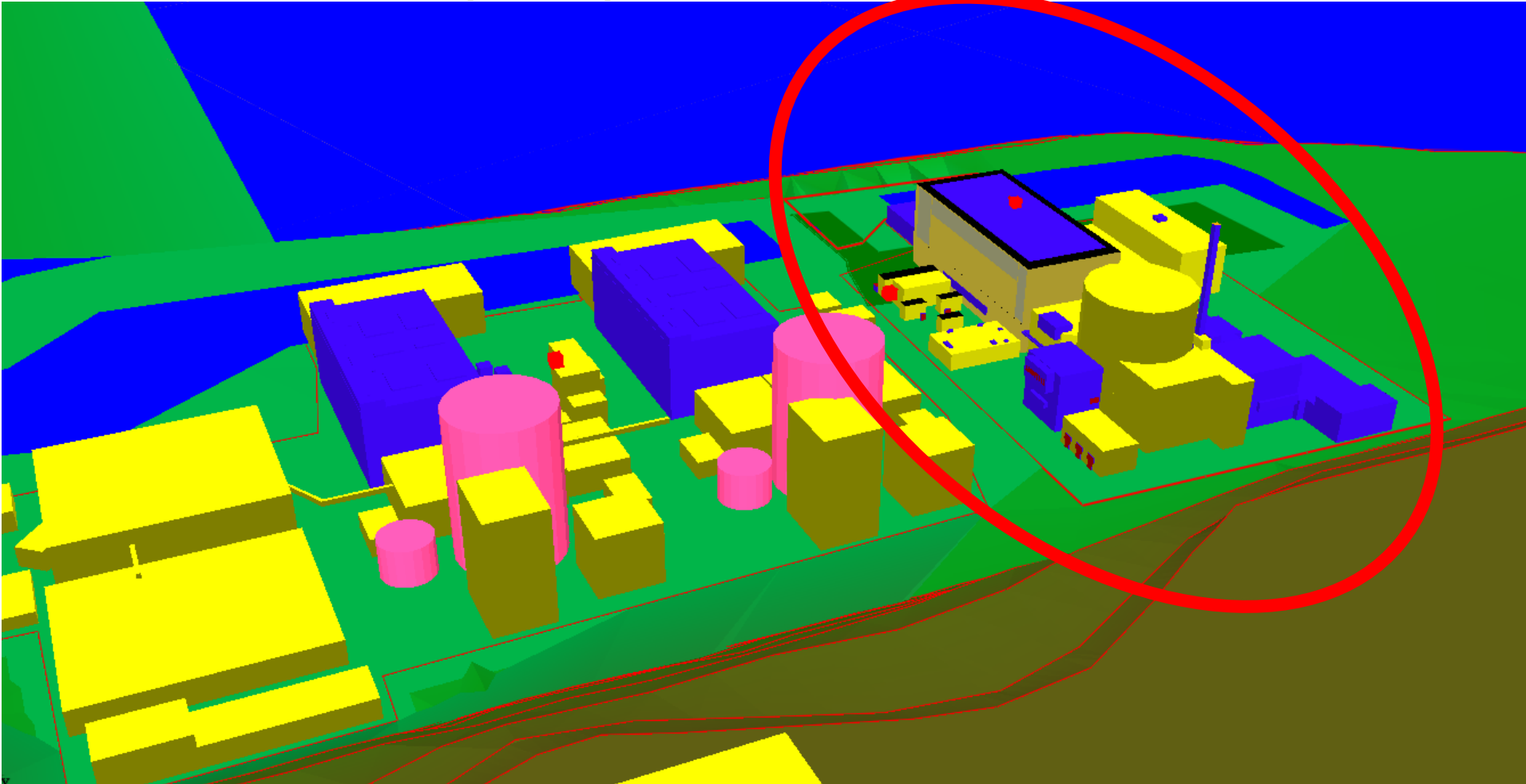
3. REAL USE CASES

- Modelling of a whole site and updating of sources



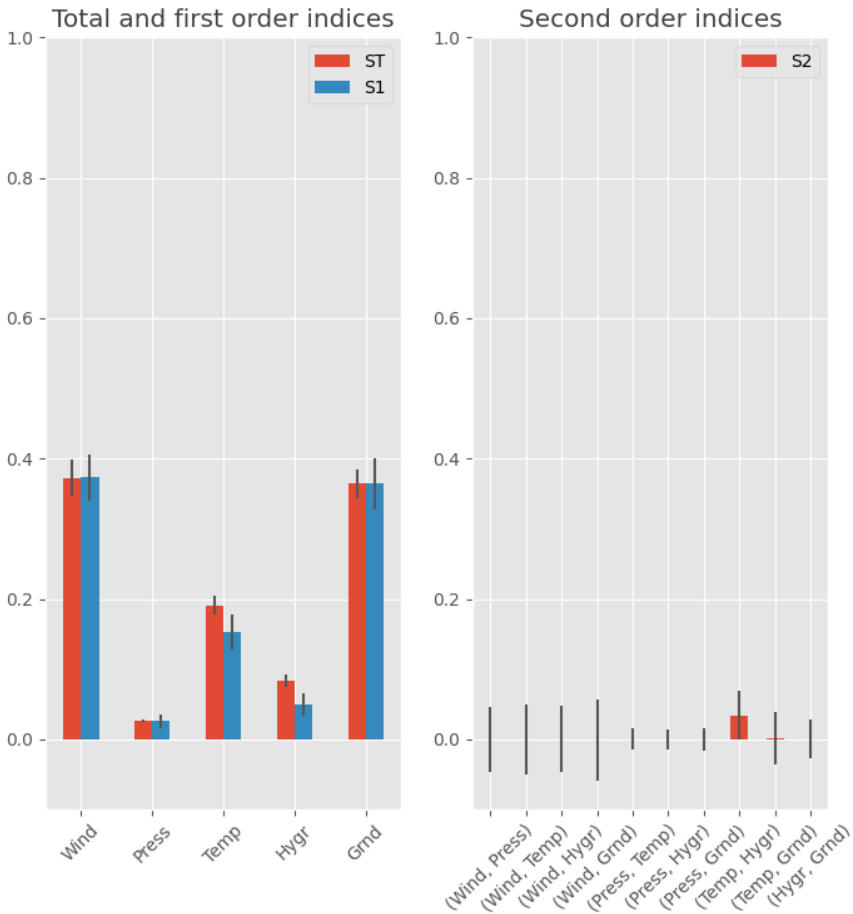
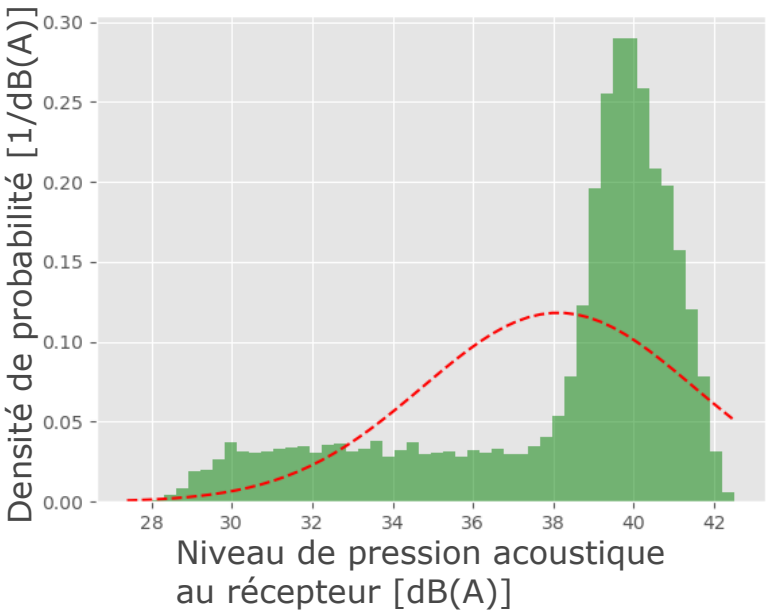
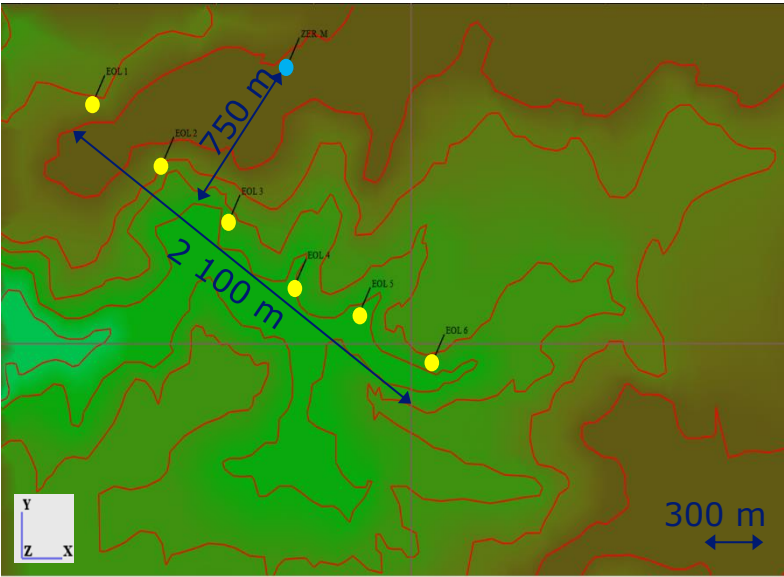
3. REAL USE CASES

- Impact of a future nuclear power plant



3. REAL USE CASES

- Propagation of uncertainties and sensitivity analysis



4. PERSPECTIVES

- A more stable and « user-friendly » software!

- Lots of improvement ideas! Priorities?
 - Ergonomics improvement to ease analysis and report writing?
 - More details on the ranking of the different sources?
 - A new engineering solver, dealing with wind effects?
 - Better sensitivity analysis and propagation of uncertainties module?
 - Library obsolescence?
 - ...

4. USEFUL LINKS

- <https://sourceforge.net/projects/code-tympan/>
- https://gitlab.com/tympan/code_tympan/-/boards
- <https://code-tympan.readthedocs.io/en/latest/index.html>

Questions ?

- codetympan@edf.fr
- <https://www.edf.fr/groupe-edf/edf-pres-de-chez-moi>